BS 161 - Cell and Molecular Biology

Spring Semester 2027

Instructor: Dr. Marcus Ellery, 214 Natural Science Building, elleryma@msu.edu. **Office hours:** Tuesdays 2:00-4:00 PM or by appointment.

Class meetings: Lectures meet Monday, Wednesday and Friday from 9:10 to 10:00 AM in 100 Natural Science. The first lecture is Monday, January 11 and the last is Friday, April 30. We do not meet during spring break (March 8-12).

Course goals

This course introduces the molecular basis of life: the structure and function of macromolecules, membranes, energy flow in cells, the flow of genetic information from DNA to protein, and how cells communicate and divide. By the end of the semester you should be able to explain how a change in a gene can change the behavior of a cell.

Exams

There are three midterm exams, each given during the regular lecture period: Exam 1 on Wednesday, February 10, Exam 2 on Wednesday, March 17, and Exam 3 on Wednesday, April 14. The cumulative final exam is on Monday, May 3 from 7:45 to 9:45 AM in our usual room. Bring a #2 pencil and your student ID to every exam.

Online quizzes

Weekly online quizzes on Mastering Biology cover the reading for the coming week. Each quiz is due at 11:59 PM on a Sunday. There are no quizzes in exam weeks or during spring break. The quiz due dates are: January 17, January 24, January 31, February 7, February 21, February 28, March 21, March 28, April 4, April 11. Your lowest quiz score is dropped.

Research paper

You will write a short research paper on a cellular process of your choice. The paper has three parts. Your topic proposal (one paragraph) is due on D2L by 5:00 PM on Friday, January 29. An annotated bibliography with at least six sources is due by 5:00 PM on Friday, February 26. The final paper (2,000 words) is due by 5:00 PM on Friday, April 23.

Group presentation

In teams of four, you will give a ten-minute presentation of your research topics during lecture on Monday, April 26. Teams will be assigned in March.

Grading

Exams 1-3	**36%**
Final exam	24%
Online quizzes	15%
Research paper	18%
Group presentation	7%

Important university dates

The last day to drop the course with a full refund is February 3. The last day to drop with no grade reported is March 3.

Attendance

Attendance is expected at every class meeting. If you must miss class for an illness or a university-sanctioned activity, email the instructor before class when possible. Repeated absences will make it difficult to succeed in this course, because much of the material is introduced only in class.

Academic integrity

Article 2.III.B.2 of the Student Rights and Responsibilities states that the student shares with the faculty the responsibility for maintaining the integrity of scholarship, grades, and professional standards. You may discuss ideas with classmates, but the work you submit must be your own. Using AI tools to write answers you submit is not allowed unless an assignment says otherwise.

Accommodations

If you have a documented disability and need accommodations, contact the Resource Center for Persons with Disabilities (RCPD) and share your Verified Individualized Services and Accommodations (VISA) form with the instructor within the first two weeks of the semester.

Late work

Late work is accepted up to 48 hours after the deadline with a 10% penalty per day. Work more than 48 hours late receives no credit unless an extension was arranged in advance. Technology problems are not an excuse, so submit early and keep copies of your files.

Communication

Announcements are posted on D2L. Check D2L and your MSU email at least once a day. Questions about the material are best asked in office hours or on the course discussion board, where classmates can benefit from the answers too.